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Application of Auction Mechanisms and Multi-Agent Technologies for Resource and Task Allocation in Distributed Computing Systems

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Abstract—The Internet of Things (IoT) integrates numerous devices with limited computational resources, where local data processing at the Edge level reduces latency and improves energy efficiency. Distributed IoT systems face challenges of scalability and adaptability, as centralized models are inefficient in dynamic environments. Multi-agent systems with auction mechanisms enable decentralized, self-organizing, and fair resource allocation, enhancing the resilience and efficiency of IoT infrastructures.

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INTRODUCTION

Distributed computing systems, including Internet of Things (IoT) infrastructures, are characterized by decentralization, heterogeneity, and high dynamism. At the core of IoT lies edge computing, which enables local data processing—critical for tasks requiring low latency, energy efficiency, and autonomy under unstable connectivity.

Existing approaches to resource management in IoT often suffer from limited scalability and an inability to adapt to changing conditions. Centralized solutions create bottlenecks and points of failure, while static algorithms fail to consider network topology and latency.

Multi-agent systems (MAS) model the computing infrastructure as a collection of autonomous agents capable of coordination, competition for resources, and adaptation to the environment, ensuring scalability and resilience.

Integrating auction mechanisms into MAS formalizes resource allocation as a process of mutual bidding among agents, where outcomes are determined by bids and auction rules. This combination of technologies fosters system self-organization, enabling fair, transparent, and efficient resource distribution.

METHODS

Resource allocation can be modeled as a virtual market [1] in which agents participate in an auction (see Fig. 1). IoT device agents act as buyers of computational resources, edge node agents as sellers, and a special auctioneer agent coordinates the entire process. This takes into account the economic motivation of participants and achieves efficient equilibrium.

Our system can be represented as a graph $G = (V, E)$, where the vertices $V = N \cup D$ include both edge nodes $N = \{n_1, n_2, \ldots n_k\}$ (nodes) and IoT devices $D = \{d_1, d_2, \ldots d_m\}$ (devices), and the edges $(d_i, n_j) \in E$ have weights $w(d_i, n_j)$ reflecting data transfer delays.

Each task from the set $T_i = \{t_{i1}, t_{i2}, \ldots t_{il_i}\}$ generated by device $d_i \in D$ is characterized by requirements for CPU $cpu(t)$, memory $mem(t)$, data volume $data(t)$, deadline $deadline(t)$ and priority $priority(t)$,

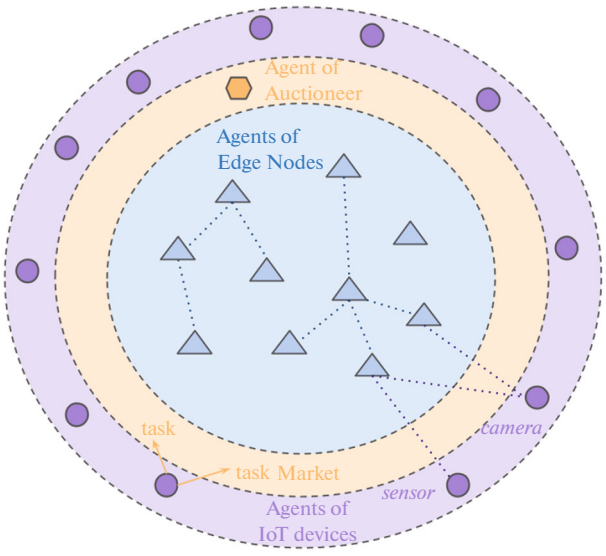


Fig. 1. High-level system design.

expressed in abstract units. Nodes $n_j \in N$ have limited resources $Cap_j = \{cpu_j, mem_j\}$, which are dynamically allocated among incoming tasks and have available capacity $Load_j$ at any time.

The auction protocol in our system works in three stages: (1) Bid submission, (2) Bid evaluation, (3) Winner determination.

On the first stage, device agents d_i form a bid $b_i(t)$ (1) for task execution:

$$b_i(t) = \max_{n_j \in N_{\text{accessible}}(d_i)} U_i(t, n_j, \tau), \quad (1)$$

where $N_{\text{accessible}}(d_i)$ is the set of nodes available for device d_i, U_i is the declared utility (value) from executing the task.

The utility function (2) [2] provides a quantitative estimate of the "benefit" or "satisfaction" a device d_i gets from executing its task on node n_j, accounting for base task value, penalty for execution delay, and energy costs:

$$U_i(t, n_j, \tau) = V_j(t)\phi(\tau, \text{deadline}(t)) - E_i(t, n_j). \quad (2)$$

The base task value (3) determines how much "benefit" a device gets from successful task execution:

$$V_j(t) = \text{priority}(t)(cpu(t) + mem(t))\beta_i, \quad (3)$$

where β_i is an importance coefficient for device d_i.

The penalty function for delay (4) shows how much the benefit is reduced by delay and grows exponentially—reflecting the critical importance of deadlines in IoT systems:

$$\phi(\tau, \text{deadline}(t)) = \begin{cases} e^{-\alpha\frac{\tau}{\text{deadline}(t)}} & \text{if } \tau \le \text{deadline}(t) \\ 0 \quad \text{if } \tau > \text{deadline}(t) \end{cases}. \quad (4)$$

Energy costs for task execution can be, for example, given by the cost of data transfer to the node and getting the result.

In the second stage, node agents calculate the cost of executing tasks C_j. The cost function (5) [3] evaluates the expenses incurred by a node n_j when executing a task from device d_i considering computational costs, communication expenses, and current load costs:

$$C_j(t, d_i) = C_{\text{comp}}(t) + C_{\text{comm}}(t, d_i) + C_{\text{load}}(t). \quad (5)$$

The computational cost (6) reflects real computational expenses:

$$C_{\text{comp}}(t) = p_{\text{cpu}} cpu(t) + p_{\text{mem}} mem(t), \quad (6)$$

where p_{cpu} and p_{mem} are the unit costs of CPU and memory.

The communication cost (7) models the "price" of data transfer:

$$C_{\text{comm}}(t, d_i) = \begin{cases} w(d_i, n_j)\, data(t)\, p_{\text{net}} & \text{if } (d_i, n_j) \in E \\ dist(d_i, n_j)\, data(t)\, p_{\text{net}}\gamma & \text{if } (d_i, n_j) \notin E \end{cases}, \quad (7)$$

where $w(d_i, n_j)$ is the direct edge weight between device and node, $dist(d_i, n_j)$ is the shortest graph distance, p_{net} is the unit cost of network traffic, $\gamma > 1$ is a penalty for indirect connection.

The load cost (8) accounts for not only direct costs but also side effects, such as performance drops under heavy load:

$$C_{\text{load}}(t) = C_{\text{comp}}(t)\left(1 + \left(Load_j / Cap_j\right)^2\right). \quad (8)$$

In stage 3, for each task t from device d_i, the auctioneer agent selects the winner node (9):

$$n^* = \arg\min_{n_j \in N_{\text{capable}}(t)} C_j(t, d_i), \quad (9)$$

where $N_{\text{capable}}(t)$ is the set of nodes able to execute task t (e.g., having sufficient resources).

Auction mechanisms aim to balance maximization of total social welfare SW (10) [4] and the interests of all auction participants:

$$SW = \sum_{i=1}^{m}\sum_{t \in T_i}\sum_{j=1}^{k} x_{i,j}^t (b_i(t) - C_j(t, d_i)), \quad (10)$$

where $x_{i,j}^t$ is an indicator that equals 1 if task t from device d_i is executed at node n_j.

The central feature of our solution is the VCG (Vickrey–Clarke–Groves) payment mechanism (11), defining how much a device should pay to ensure that agents have no incentive to lie about their value/cost:

$$p_i^{VCG}(t) = SW_{-i} - \left(SW - v_i\left(x_i^*\right)\right), \quad (11)$$

where SW_{-i} is the optimal social welfare without device d_i, $v_i\left(x_i^*\right)$ is the value obtained by agent d_i at the optimal allocation.

The VCG mechanism ensures the truth-revealing property—it is profitable for participants to honestly declare their preferences. Furthermore, VCG maximizes social welfare and ensures individual rationality of participation.

RESULTS AND DISCUSSION

For system demonstration, a city traffic optimization scenario was considered, where IoT devices detect events, and Edge nodes perform preprocessing, accident recognition, and congestion prediction for automated traffic light control. Efficiency was evaluated using four metrics: social welfare, resource utili-

Table 1. Scalability analysis

Setup size	Welfare, %	Efficiency, %	Fairness	Time, ms
Small	89.2	91.5	0.82	45
Medium	91.8	88.7	0.84	78
Large	90.3	86.4	0.81	175

zation, fairness according to Jain's index, and convergence time. Experiments were conducted on small (10 nodes, 20 devices), medium (25 nodes, 50 devices), and large (50 nodes, 100 devices) scale testbeds with parameters calibrated to real equipment characteristics.

The results demonstrate good system scalability: as the number of nodes increases from 10 to 50, efficiency remains at the level of 85–90%, and convergence time grows linearly. This confirms the practical applicability of the proposed approach.

CONCLUSIONS

The decentralized approach to resource allocation in edge computing has been developed, combining theoretical optimality with practical scalability. Achievements include an architecture without a single point of failure, consideration of network topology and latency, as well as guarantees of optimality and resilience during scaling. Promising directions include the development of hybrid auction schemes and the transition to industrial implementations.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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